THANATHAI LERTPETCHPUN

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RESEARCH INTEREST

Controllable Speech Generation, Speech Emotion Recognition, Privacy-related Speech Technology

EDUCATION

PhD in Electrical and Computer Engineering, University of Southern California (USC) 2024 - Present
Advisor: **Prof. Shrikanth Narayanan**
Current GPA: 3.87/4.00

Master of Science in Electrical and Computer Engineering, USC 2024 - 2026
Advisor: **Prof. Shrikanth Narayanan**
GPA: 3.87/4.00

Bachelor in Computer Engineering, Chulalongkorn University 2018 - 2022
Capstone Project Advisor: **Prof. Ekapol Chuangsuwanich**
GPA: 3.82/4.00, CS-only 4.00/4.00

WORK EXPERIENCE

Graduate Research Assistant 08/2025 - Present
Signal Analysis and Interpretation Labortory (SAIL) *Los Angeles, CA*

- Develop learning-free and controllable frameworks for accented, multilingual text-to-speech.
- Build state-of-the-art speech emotion recognition systems for attribute and categorical emotion prediction.
- Propose a speaker-poisoning method to erase voice-cloning capability in zero-shot text-to-speech.

Undergrad Research and Teaching Assistant 05/2021 - 08/2024
Chulalongkorn University *Bangkok, Thailand*

- Taught undergraduate students Computer Programming, Discrete Mathematics, Linear Algebra, Algorithm Design, and Pattern Recognition.

Data Scientist 06/2021 - 07/2021
Tencent (Thailand) *Bangkok, Thailand*

- Built an end-to-end multi-GPU ML pipeline for news recommendation.
- Designed and implemented an ML architecture for music recommendation.

PROJECTS

Annonymous Real Time Speech (USC - JHU - Meaning Company) 10/2024 - Present

- Developing learning-free framework for multilingual TTS on British, Spanish, and Indian Accent.
- Proposed Accent Vector, a controllable accent manipulation method for multilingual TTS that requires no accented training data.
- Proposed a speaker poisoning framework that erases a target speaker's voice-cloning capability in zero-shot TTS to protect against unauthorized voice generation. (1 COLM ReData Workshop 2026 publication)
- Analyzed the effect of speaker embedding on accent strength using phonemes rules to control the effect. (1 ICASSP2026 publication)
- Proposed frameworks to develop state-of-the-art speech emotion recognition (SER) for predicting both emotional attributes and categorical emotions. They achieved 1st rank in emotional attributes and 2nd rank in categorical attributes in Naturalistic Conditions Challenge (2 Interspeech2025 publications)

Speaker Recognition System and Anti-spoofing System. (Chulalongkorn University) 08/2022 - 05/2024

- Proposed an approach to amplify artifacts in spoofed utterances using speech enhancement and show its robustness on different countermeasures and speech enhancement models (1 Interspeech2025 publication).
- Proposed a new normalization layer to address the problem of mismatched emotions and languages in speaker verification. (1 Interspeech2023 publication).

SELECTED PUBLICATIONS

(COLM Re-Data Workshop, 2026) **Thanathai Lertpetchpun***, Thanapat Trachu*, Sai Praneeth Karimireddy, Shrikanth Narayanan. "Speech Generation Speaker Poisoning: Capability Erasure in Zero-Shot Text-to-Speech"

(ACM KDD, 2026) Tiantian Feng, Kevin Huang, Anfeng Xu, Xuan Shi, **Thanathai Lertpetchpun**, Jihwan Lee, Yoonjeong Lee, Dani Byrd, Shrikanth Narayanan. "Voxlect: A Speech Foundation Model Benchmark for Modeling Dialects and Regional Languages Around the Globe"

(ICASSP, 2026) **Thanathai Lertpetchpun***, Yoonjeong Lee*, Thanapat Trachu, Jihwan Lee, Tiantian Feng, Dani Byrd, Shrikanth Narayanan. "Quantifying Speaker Embedding Phonological Rule Interactions in Accented Speech Synthesis"

(Interspeech, 2025) **Thanathai Lertpetchpun***, Tiantian Feng*, Dani Byrd, Shrikanth Narayanan. "Developing a High-performance Framework for Speech Emotion Recognition in Naturalistic Conditions Challenge for Emotional Attribute Prediction."

(Interspeech, 2025) Tiantian Feng*, **Thanathai Lertpetchpun***, Dani Byrd, Shrikanth Narayanan. "Developing A Top-tier Framework in Naturalistic Conditions Challenge for Categorized Emotion Prediction: From Speech Foundation Models and Learning Objective to Data Augmentation and Engineering Choices."

(Interspeech, 2025) Thanapat Trachu*, **Thanathai Lertpetchpun***, Ekapol Chuangsuwanich. "Amplifying Artifacts with Speech Enhancement in Voice Anti-spoofing."

(Interspeech, 2023) **Thanathai Lertpetchpun**, Ekapol Chuangsuwanich. "Instance-based Temporal Normalization for Speaker Verification."

(* denotes equal contribution)

AWARDS AND HONORS

ICASSP 2026 Travel Grant Student Travel Grant for ICASSP 2026
SER in Naturalistic Conditions Challenge (2025) Rank 1st in Attribute Prediction
SER in Naturalistic Conditions Challenge (2025) Rank 2nd in Categorical Prediction
Viterbi School of Engineering Fellowship (2024) Top-off fellowship selected among incoming Ph.D. students
Interspeech 2023 Travel Grant Student Travel Grant for Interspeech2023
Thailand Olympiad Informatics (2017) Ranked in the top-15 among all Thai high school contestants

SKILLS

Languages: Thai (Native), English (Fluent).
Computer Languages: Python, C/C++, Bash Script, JavaScript, Node.js.
Framework: Pytorch, Espnet, Huggingface, Speechbrain.

SERVICE

Reviewer: Interspeech 2026, SLT 2026, COLM ReData Workshop 2026
Social Chair at SAIL (2026 - Present)
Deputy Social Chair at SAIL (2025)
Military Service (2015-2018)